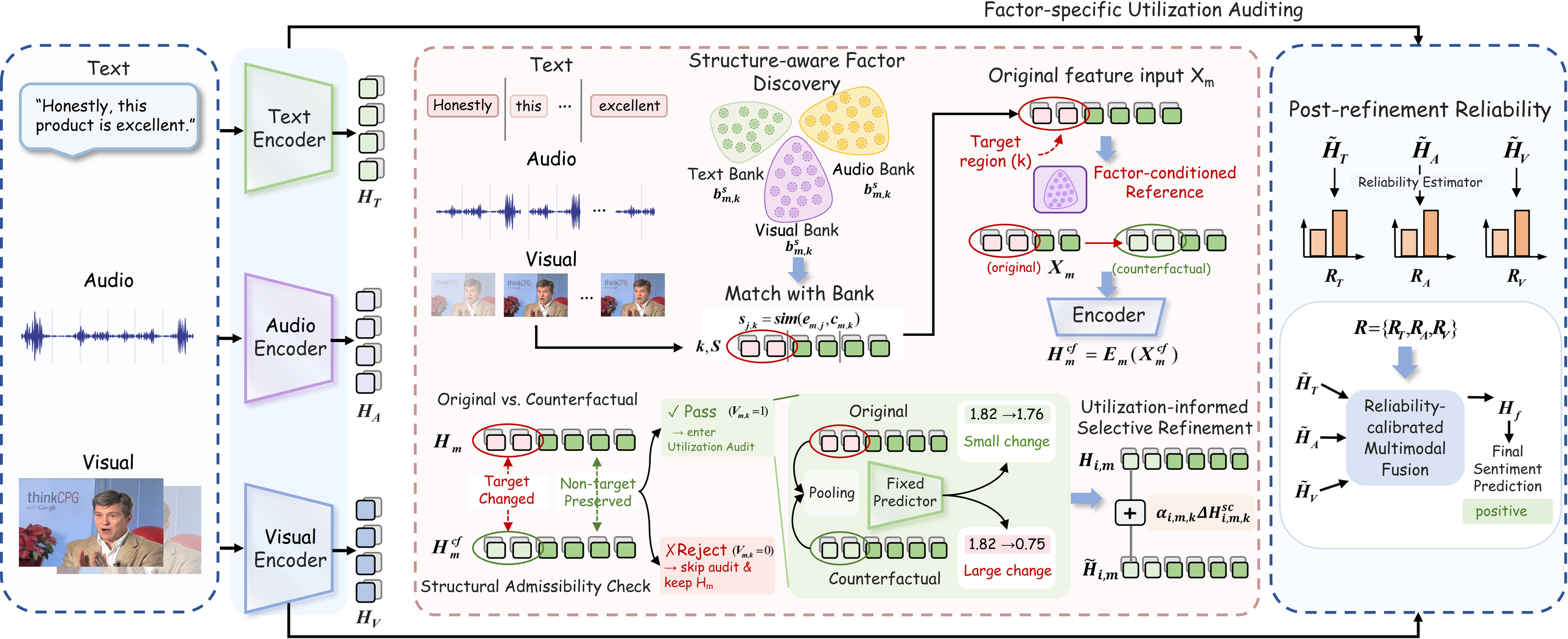

Factor-specific Utilization Auditing
Text
"Honestly, this product is excellent."
Audio
Visual
Text Encoder
Audio Encoder
Visual Encoder
H_T
H_A
H_V
Text
Honestly
this
...
excellent
Audio
Visual
Structure-aware Factor Discovery
Text Bank $b^s_{m,k}$
Audio Bank $b^s_{m,k}$
Visual Bank $b^s_{m,k}$
Match with Bank
$s_{j,k} = sim(e_{m,j}, c_{m,k})$
k, S
Original feature input X_m
Target region (k)
Factor-conditioned Reference
(original) X_m
(counterfactual)
Encoder
$H^{cf}_m = E_m(X^{cf}_m)$
Original vs. Counterfactual
H_m
Target Changed
Non-target Preserved
H^{cf}_m
Structural Admissibility Check
✓ Pass ($V_{m,k}=1$)
→ enter Utilization Audit
✗ Reject ($V_{m,k}=0$)
→ skip audit & keep H_m
Original
Pooling
Fixed Predictor
Counterfactual
1.82 →1.76
Small change
1.82 →0.75
Large change
Utilization-informed Selective Refinement
$H_{i,m}$
$\alpha_{i,m,k} \Delta H^{sc}_{i,m,k}$
$\tilde{H}_{i,m}$
Post-refinement Reliability
$\tilde{H}_T$
$\tilde{H}_A$
$\tilde{H}_V$
Reliability Estimator
R_T
R_A
R_V
$R=\{R_T, R_A, R_V\}$
Reliability-calibrated Multimodal Fusion
H_f
Final Sentiment Prediction
positive